control would have a hard time resisting the temptation to sneak a peek: as the electron or photon passes through the doubly slit screen or the beam splitter, why not take a quick look to see what path it *really* follows on its way to the detector? In the double-slit experiment, why not put little detectors in front of each slit to tell you whether the electron went through one opening, the other, or both (while still allowing the electron to carry on toward the main detector)? In the beam-splitter experiment, why not put, on each pathway leading from the beam splitter, a little detector that will tell if the photon took the left route, the right route, or both routes (again, while allowing the photon to keep going onward toward the detector)?

The answer is that you *can* insert these additional detectors, but if you do, you will find two things. First, each electron and each photon will always be found to go through one and only one of the detectors; that is, you can determine which path each electron or photon follows, and you will find that it always goes one way or the other, not both. Second, you will also find that the resulting data recorded by the main detectors have changed. Instead of getting the interference patterns of Figure 4.3b and 7.1b, you get the results expected from classical physics, as in Figure 4.3a. By introducing new elements—the new detectors—you have inadvertently changed the experiments. And the change is such that the paradox you were *just* about to reveal—that you now know which path each particle took, so how could there be any interference with another path that the particle demonstrably did not take?—is averted. The reason follows immediately from the last section. Your new observation singles out those histories that could have preceded whatever your new observation revealed. And since this observation determined which path the photon took, *we consider only those histories that traverse this path, thus eliminating the possibility of interference.*

Niels Bohr liked to summarize such things using his *principle of complementarity.* Every electron, every photon, every*thing*, in fact, has both wavelike and particlelike aspects. They are complementary features. Thinking purely in the conventional particle framework—in which particles move along single, unique trajectories—is incomplete, because it misses the wavelike aspects demonstrated by interference patterns.[16] Thinking purely in the wavelike framework is incomplete, because it